Reworked SMB inference phase

ContinuIX, GROUP_UNIL method01

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The first run of the SMB inference stopped after 10–20 L-BFGS iterations and reduced the loss by a few percent. We reran *only* the SMB phase of all 96 experiment–glacier pairs on the saved data-assimilation (DA) results. The DA, the emulator, the SMB profile ($\Delta z = 200\,\mathrm{m}$), its regularisation (0.05) and the stopping tolerances are unchanged. The design was tuned on the synthetic glacier S01, where the true SMB is known, and then applied to everything.

What changed

1. **Domain border.** The 5-cell ramp towards zero thickness at the domain edge now acts only outside the ice mask. Before, it removed ice from inside the glacier at every step (every glacier except G01 touches the edge).
2. **Fixed time step.** One step, from the CFL condition of the initial state, ending exactly at t_2. Before, the step followed the speed and overshot t_2, so the loss jumped by $\sim$0.05 m under tiny changes and the line search failed.
3. **Cheaper line search.** Trial steps run the forward model only; value and slope at the current point are reused.
4. **Band misfit.** The loss is the RMS of the mean thickness residual in each 50 m elevation band (before: per-cell RMSE). An SMB profile can only change band totals, not the redistribution of ice inside a band, which comes from the DA thickness.
5. **Geodetic term.** The glacier-wide mean SMB is tied to the observed mean thickness change (weight 100).

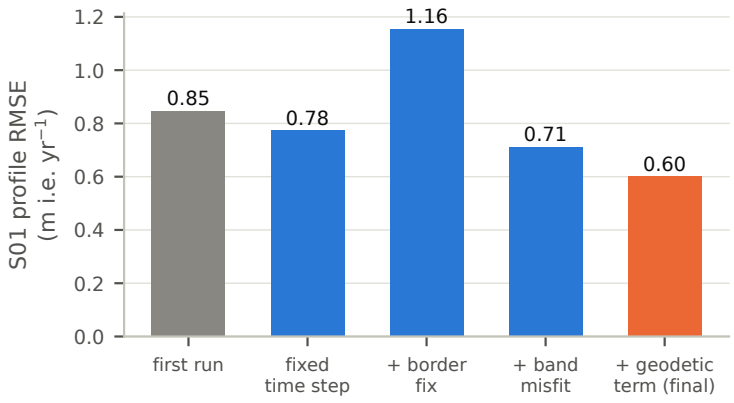


Figure 1: S01 (EXP01): family-mean profile error against the prescribed law, adding the changes one by one. The border fix alone worsens S01, because the old ramp happened to remove over-thick DA ice at the glacier head; the band misfit and the geodetic term give the improvement.

SMB-inference loss, before and after

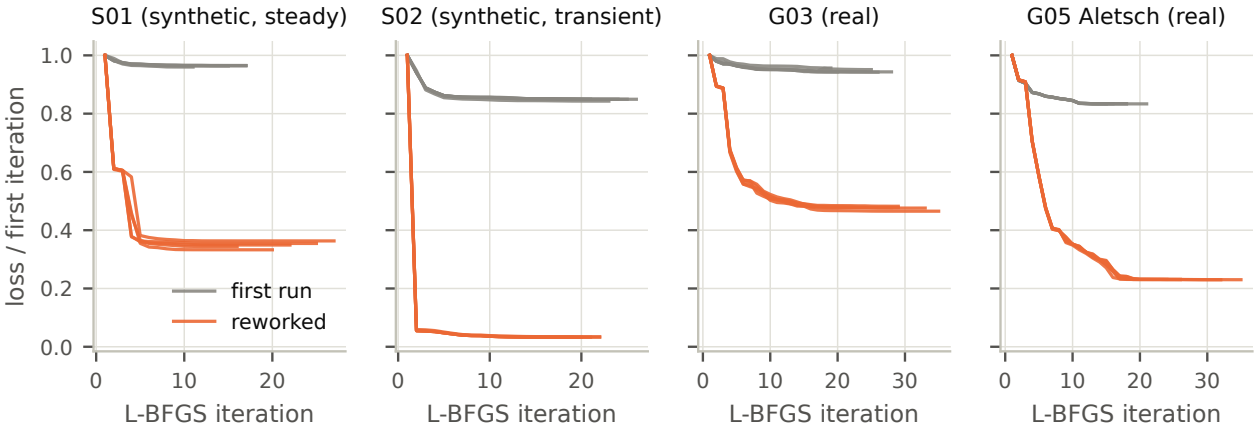


Figure 2: Loss of every family member, relative to its first iteration. The old loss flattens at 83–97 % of its start; the reworked loss keeps decreasing.

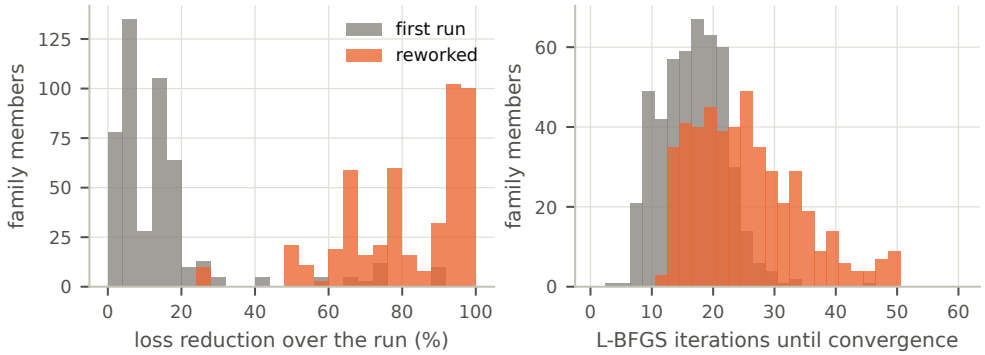


Figure 3: All 478 family members. Median loss reduction 12 % → 85 %; 20 % of the old runs reduced the loss by less than 5 %, none of the new ones. Median iterations 17 → 24. The two losses are not the same quantity (cell RMSE vs. band misfit plus geodetic term), so Fig. 4 gives a common yardstick.

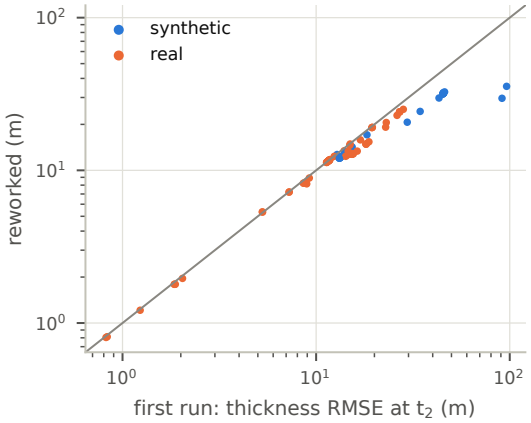


Figure 4: Per-cell thickness error at t_2 (not what the new loss optimises) for the 96 pairs: lower in 94; median 14.5 m → 12.7 m.

Results

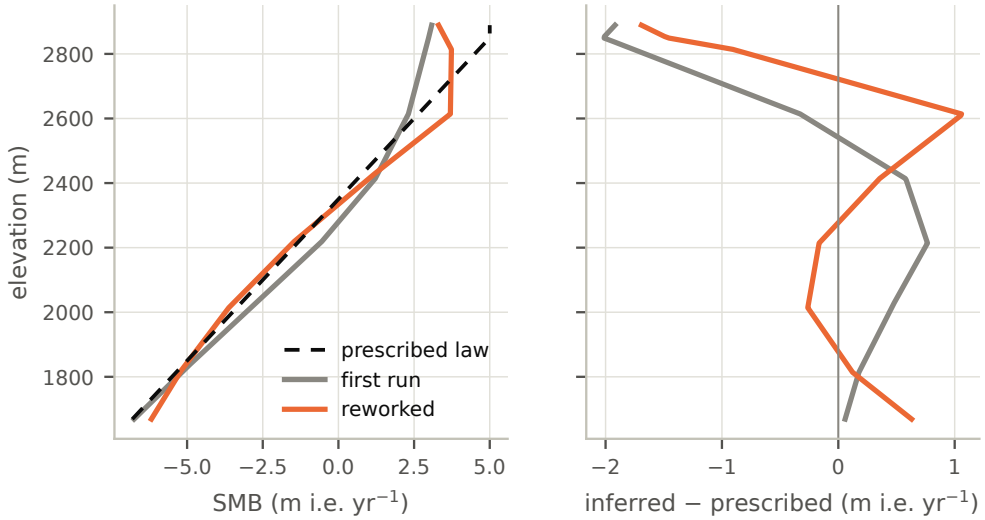


Figure 5: S01 (EXP01), family-mean SMB profile against the prescribed law (RMSE $0.85 \rightarrow 0.60$ m i.e. yr^{-1}).



Figure 6: All S01 experiments: profile RMSE against the prescribed law. Better in 12 of 14; worse in EXP05. In EXP05 the thickness perturbation puts ice below the real terminus, so the ice mask reaches 230 m lower and the glacier-wide constraint follows that inconsistent data.

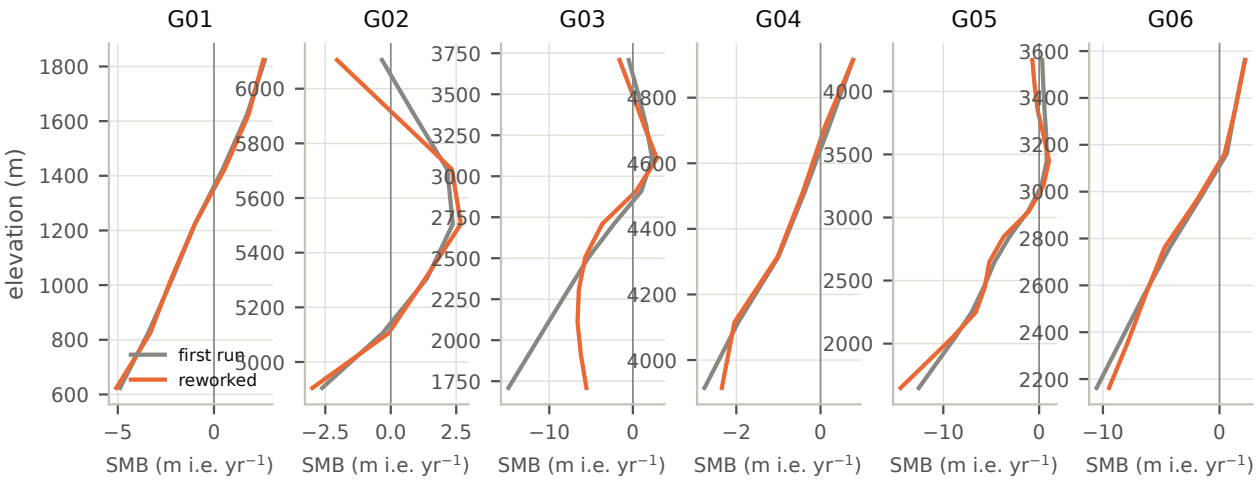


Figure 7: Real glaciers (EXP01), family-mean SMB profiles. Most are close to the first run; G03 changes most (its terminus touches the domain edge, where the old ramp removed ice).

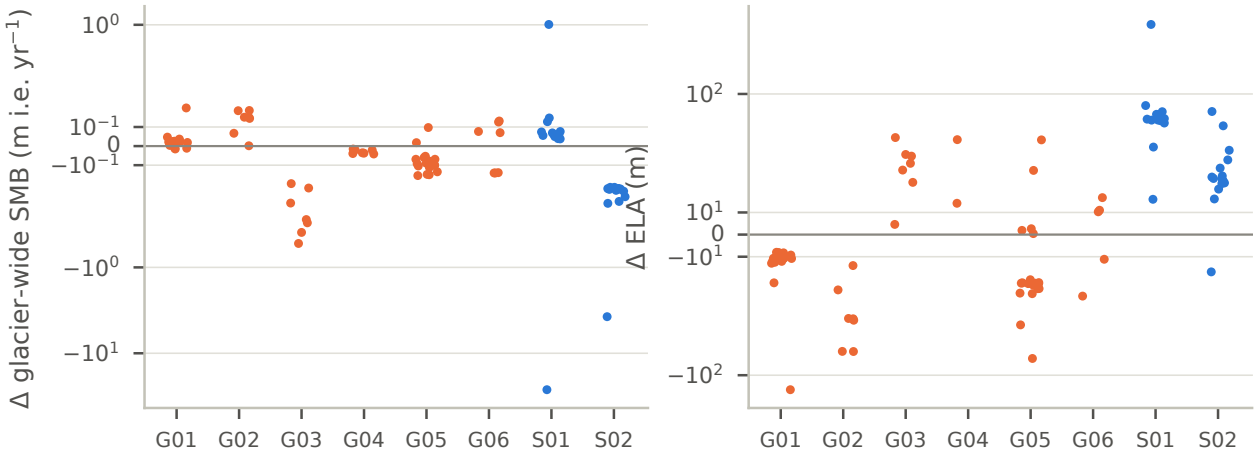


Figure 8: Change (reworked minus first run) of glacier-wide SMB and ELA, one dot per experiment (symmetric-log axes). Median $|\Delta\text{SMB}|$ 0.09 m i.e. yr^{-1} (real glaciers 0.08); median $|\Delta\text{ELA}|$ 24 m. The large S01 changes are EXP15, whose coarse grid had failed in the first run (glacier-wide SMB error +27 m i.e. yr^{-1}). Median σ_{SMB}: 2.27 → 2.02 m i.e. yr^{-1}.

Remaining limits

- S01 is not fitted exactly. With model-generated data the inversion recovers the law (RMSE 0.05); with the observed data the error comes from the DA state (ice volume 1.7–1.9× the truth, speed too low), not from the optimiser, regularisation or profile resolution.
- The fixed step is chosen from the initial speeds. In one member (EXP20 G01, $\tau_{\text{ref}} = 1.79$) the run diverged; it was rerun with a halved CFL number (0.1).
- The new results are in `results/<EXP>_<GID>_cnn/smb_G_bands_geod/export/`; `GROUP_UNIL/` still holds the first run.